Calculations for Beam at Focal Plane and Controlled Defocus

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2026

1 Overview

This document provides the core equations needed to (i) distinguish focal length from working distance, (ii) predict the beam diameter at any axial position around focus, (iii) solve the inverse problem—finding the axial position at which the beam reaches a desired diameter, and (iv) evaluate reference cases for beam-diameter growth of $2\%, 10\%, 15\%, 30\%, 50\%, 100\%$, and 200%. Two beam types are treated: the **Gaussian beam** and the **flat-top beam**. All results follow Siegman [1], Saleh & Teich [2], Kogelnik & Li [3], Goodman [4], Smith [5], Shealy & Hoffnagle [6], and Harris et al. [7].

2 Focal Length vs. Working Distance

Focal length f. Distance from the rear *principal plane* of the focusing system to the focal point. Optical quantity fixed by lens design [5, 4]. **This is the value used in every equation below. Working distance WD.** Distance from the last *physical* surface of the lens (or the manufacturer-defined mechanical reference) to the focal plane. Mechanical mounting number. For a compound lens, $f \neq$ WD in general. Use f in the equations; use WD only to physically position the workpiece.

3 The f-θ Scan Lens

$$x \;=\; f\,\theta \tag{1}$$

Purpose. Predicts the focused spot position on the workpiece from the scan angle [7]. **Variables.** x = lateral displacement; f = focal length; θ = scan angle in radians.

4 Gaussian Beam Model

4.1 Intensity distribution

$$I(r,z) \;=\; I_0(z)\,\exp\!\left[-\frac{2r^2}{w^2(z)}\right] \tag{2}$$

Purpose. Radial irradiance profile [1, 2]. **Variables.** $I(r,z)$ = irradiance; $I_0(z)$ = on-axis peak; $w(z) = 1/e^2$ radius; r = radial coordinate.

4.2 Focused beam waist after a lens

$$w_0 \;=\; \frac{M^2\,\lambda\,f}{\pi\,w_L} \;=\; \frac{2\,M^2\,\lambda\,f}{\pi\,D} \tag{3}$$

Purpose. Minimum beam radius at focus [1, 2]. **Variables.** w_0 = focused $1/e^2$ radius; λ = wavelength; f = focal length; w_L = incident $1/e^2$ radius at lens; $D = 2w_L$; M^2 = beam quality factor.

$$d_0 \;=\; 2\,w_0 \;=\; \frac{4\,M^2\,\lambda\,f}{\pi\,D} \tag{4}$$

Purpose. Focal-plane diameter form of Eq. (3).

4.3 Rayleigh range

$$z_R \;=\; \frac{\pi\, w_0^2}{M^2\, \lambda} \tag{5}$$

Purpose. Axial distance from waist at which cross-sectional area doubles [3, 1]. **Variables.** z_R = Rayleigh range; $w_0,\ M^2,\ \lambda$ as above.

5 Forward Problem: Beam Diameter at a Chosen Axial Position

$$\boxed{d(z) \;=\; d_0 \sqrt{1 + \left(\frac{z - z_0}{z_R}\right)^2}} \tag{6}$$

Purpose. Predicts beam diameter at any axial position (forward problem) [1, 2]. **Variables.** $d(z)$ = beam diameter at position z; d_0 = focal-plane diameter (Eq. (4)); $z - z_0$ = signed axial offset from focus; z_R = Rayleigh range.

6 Inverse Problem: Axial Position for a Desired Beam Diameter

Setting $d(z) = d_{\text{target}}$ in Eq. (6) and solving:

$$\boxed{\Delta z \;=\; z - z_0 \;=\; \pm\, z_R \sqrt{\left(\frac{d_{\text{target}}}{d_0}\right)^2 - 1}} \tag{7}$$

Purpose. How far from focus to place the workpiece to get a specific spot size. **Variables.** Δz = distance from focal plane; d_{target} = desired diameter; d_0 = focused diameter; z_R = Rayleigh range. The $\pm$ gives symmetric planes before/after focus.

6.1 Percentage-increase form

Let $\varepsilon = d_{\text{target}}/d_0 - 1$ be the fractional diameter increase.

$$\boxed{\Delta z_\varepsilon \;=\; \pm\, z_R \sqrt{(1+\varepsilon)^2 - 1}} \tag{8}$$

Purpose. Compact form for standard defocus targets. **Variables.** ε = fractional diameter increase ($\varepsilon = 0.15$ means +15%).

7 Flat-Top Beam Model

Flat-top profiles are analyzed by Shealy & Hoffnagle [6].

7.1 Diffraction-limited (Airy) model

$$r_{\text{Airy}} \;=\; 1.22\,\frac{\lambda f}{D}, \qquad d_{\text{Airy}} \;=\; 2.44\,\frac{\lambda f}{D} \tag{9}$$

Purpose. Diffraction-limited focal spot for a uniform circular pupil of diameter D [4]. **Variables.** r_{Airy} = first-zero radius; d_{Airy} = first-zero diameter. **Caution.** Not a Gaussian $1/e^2$ definition.

7.2 Measured-caustic (second-moment) model

$$w_\sigma(z) \;=\; w_{\sigma,0}\sqrt{1+\left(\frac{z-z_0}{z_{R,\sigma}}\right)^2}, \qquad z_{R,\sigma} \;=\; \frac{\pi\, w_{\sigma,0}^2}{M^2\lambda} \tag{10}$$

Purpose. Data-driven propagation model for the delivered beam [1]. **Variables.** $w_{\sigma,0}$ = second-moment waist radius; $z_{R,\sigma}$ = second-moment Rayleigh range.

$$d_\sigma(z) \;=\; d_{\sigma,0}\sqrt{1+\left(\frac{z-z_0}{z_{R,\sigma}}\right)^2}, \qquad \Delta z \;=\; \pm\, z_{R,\sigma}\sqrt{\left(\frac{d_\sigma}{d_{\sigma,0}}\right)^2-1} \tag{11}$$

Purpose. Forward and inverse problems for a flat-top beam using second-moment diameters $d_\sigma = 2w_\sigma$.

8 Example: 200 W Fiber Laser with f-θ Lens ($f = 163\,\mathrm{mm}$, WD = $192\,\mathrm{mm}$)

Use $f = 163\,\mathrm{mm}$ in Eqs. (3) and (4); place the workpiece near WD = $192\,\mathrm{mm}$ and confirm the true focal plane experimentally. Reading multipliers from Table 1:

$$\Delta z_{2\%} = \pm 0.201\, z_R, \quad \Delta z_{10\%} = \pm 0.458\, z_R, \quad \Delta z_{15\%} = \pm 0.568\, z_R,$$

$$\Delta z_{30\%} = \pm 0.831\, z_R, \quad \Delta z_{50\%} = \pm 1.118\, z_R, \quad \Delta z_{100\%} = \pm 1.732\, z_R, \quad \Delta z_{200\%} = \pm 2.828\, z_R.$$

9 Reference Calculations

Diameter increase	ε	$(1+\varepsilon)^2-1$	$\sqrt{(1+\varepsilon)^2-1}$	**Peak-irradiance retention**
+2%	0.02	0.0404	0.2010	96.1%
+10%	0.10	0.2100	0.4583	82.6%
+15%	0.15	0.3225	0.5679	75.6%
+30%	0.30	0.6900	0.8307	59.2%
+50%	0.50	1.2500	1.1180	44.4%
+100%	1.00	3.0000	1.7321	25.0%
+200%	2.00	8.0000	2.8284	11.1%

Table 1: Axial defocus multipliers (Eq. (8)) and Gaussian peak-irradiance retention at constant power.

10 Assumptions and Limitations

- Paraxial propagation, negligible aberrations, homogeneous medium.
- The Gaussian $w(z)$ law describes real beams only through the M^2-embedded form of Eq. (6).
- Airy equations require an ideal uniform pupil.
- At constant power, Gaussian peak irradiance scales as $1/d^2$.

References

[1] A. E. Siegman, *Lasers*, University Science Books, 1986.

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[7] Z. B. Harris et al., "Design and characterization of telecentric f-θ scanning lenses for broadband terahertz frequency systems," *AIP Advances*, vol. 10, 125313, 2020.